Curs 5 – Summarizing Measured Data

Overview

- Ratio games
- Summarizing measured data

Ratio games

Example

- Mice trials of 2 types of cancer medicine: **M1** and **M2**.
    - Stats for the last 1000 mices.

	M1	M2
1000	**900**	800
Advanced Stage	100/30	400/210
	30%	**52.5%**
Early Stage	900/870	600/590
	96.6%	**98.3%**

Example

Bench-mark	System 6502	System 8080
Block	41.16	51.50
Sieve	63.17	48.08
Sum	104.33	99.58
Avg	52.17	49.79

System 6502	System 8080
1.00	1.25
1.00	0.76
2.00	2.01
1.00	1.01

System 6502	System 8080
0.80	1.00
1.31	1.00
2.11	2.00
1.06	1.00

- Ratio of totals
    - 6502 is worse - takes 4.7% more time
- 6502 as base
    - 6502 is better - takes 1% less time
- 8080 as base
    - 6502 is worse - takes 6% more time

System	Throughput	Response	Power
A	10	2	5
B	4	1	4

- Throughput: A is better
- Response: A is worse
- Power: A is better

Alternative	Without	With	Ratio
A on X	2	4	2.00
B on Y	3	5	1.66

- Incomparable bases: need to try both on the same machine

System A:	Test	Total	Pass	% Pass
	1	300	60	20%
	2	50	2	4%
	Total	350	62	20.6%
System B:	Test	Total	Pass	% Pass
	1	32	8	25%
	2	500	40	8%
	Total	532	48	9%

- System B is better on both tests
- System A is better overall

Guidelines

1. System better on all benchmarks
    - contradicting conclusions cannot be drawn by any ratio game technique
2. System better on all benchmarks
    - a better relative performance can be shown by selecting appropriate base
3. System better on some benchmarks
    - contradicting conclusions can be drawn in some cases (not in all cases)
4. Performance metric is an LB metric
    - better to use your system as the base
5. Performance metric is an HB metric
    - better to use your opponent as the base

6. Benchmarks that perform better on your system should be elongated and those that perform worse should be shortened

Takeaway

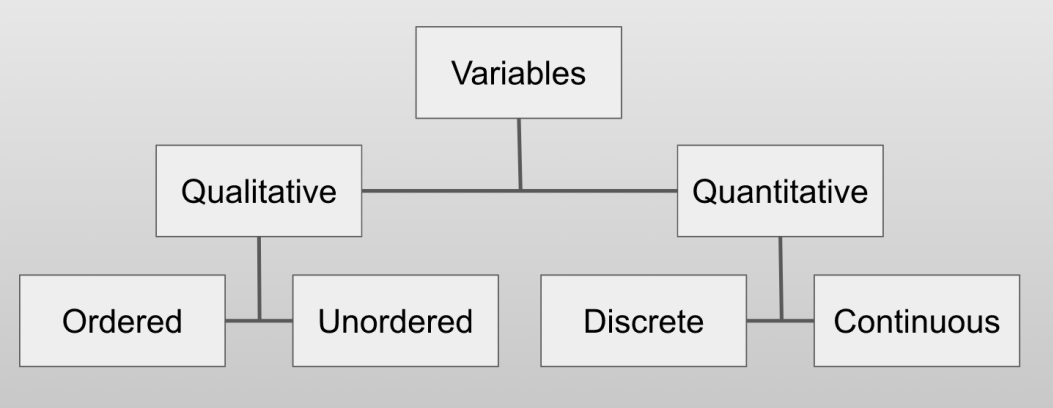
- Ratio games arise from the use of incomparable bases

Summarising measured data

Overview

- Basic probability and statistics concepts: CDF, PDF, PMF, Mean, Variance, CoV, Normal distribution
- Summarising data by a single number: mean, median, mode, arithmetic/geometric/harmonic means
- Summarising variability: range, variance, percentiles, quartiles

Recap: Types of variables



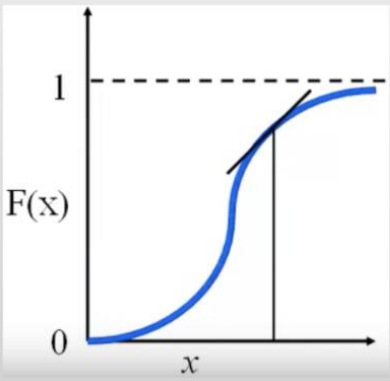
Basic concepts

- Independent events
  - The occurence of one event does not affect the probability of the other event.
  - Examples
    - Successive throws of a coin - independent
    - Price of a stock on successive days – dependent
- Random variable
  - Takes one of a specified set of values with a specified probability.
  - Examples
    - Temperature
    - Execution time

CDF, PDF, PMF

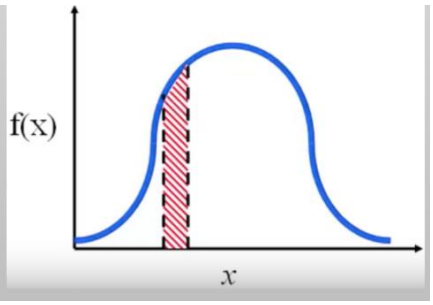
- Cumulative Distribution Function
    - Starts at 0, ends at 1

$$F(x) = P(X \leq x)$$
$$F_x(a) = P(x \leq a)$$



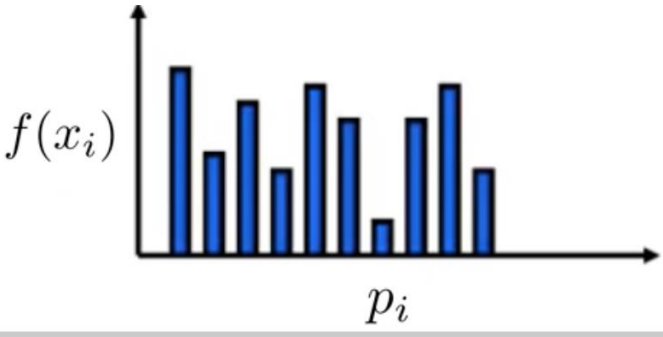
- Probability Density Function
    - Starts at 0, ends at 0.

$$f(x) = \frac{dF(x)}{dx}$$
$$P(a < x < b) = F(b) - F(a) = \int_a^b f(x)dx$$



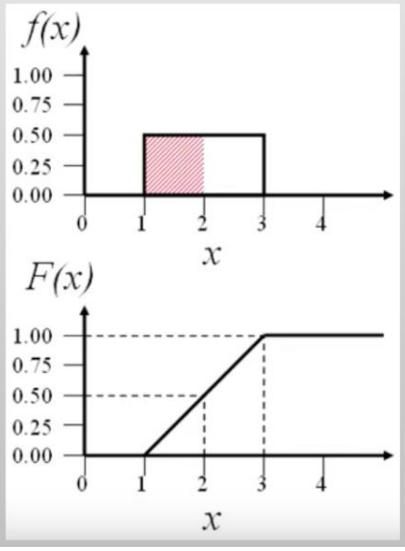
- Probability Mass Function
    - For discrete random variables

$$f(x) = P(X = x)$$
$$f(x_i) = p_i$$



Example

- Response time – uniformly distributed between 1 and 3 seconds



$$f(x) = \frac{1}{2}, \quad 1 \leq x \leq 3$$

$$P(x \leq 2) = 0.5$$

$$F(x) = \int_{-\infty}^{x} f(x)dx$$

Mean, Variance, CoV

- Mean

$$\mu = E(x) = \sum_{i=1}^{n} p_i x_i = \int_{-\infty}^{+\infty} x f(x)dx$$

- Variance
  - Expected value of the square of the distance between x and its mean.

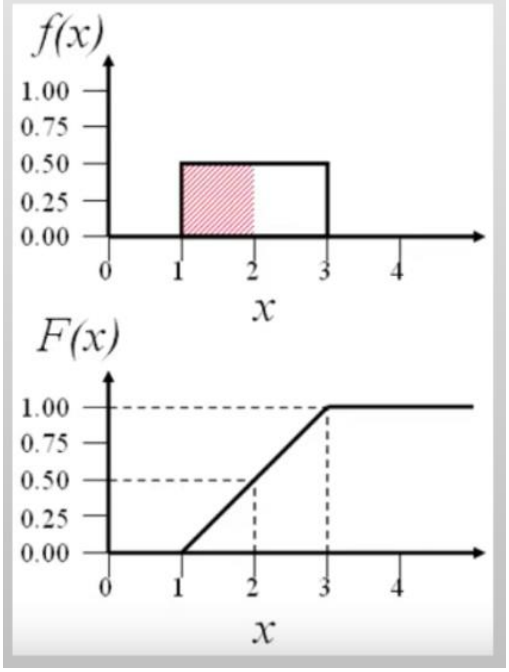
$$Var(x) = \sigma^2 = E[(x-\mu)^2] = \sum_{i=1}^{n} p_i (x_i - \mu)^2 = \int_{-\infty}^{+\infty} (x_i - \mu)^2 f(x)dx$$

- Coefficient of Variation

$$C.O.V. = \frac{Standard Deviation}{Mean} = \frac{\sigma}{\mu}$$

Example

$$f(x) = \frac{1}{2}, \quad 1 \le x \le 3$$



- **Mean**

$$\mu = E(x) = \int_{-\infty}^{+\infty} x f(x) dx = \int_{1}^{3} x \frac{1}{2} dx = \frac{x^2}{4} = \frac{9}{4} - \frac{1}{4} = 2$$

- **Variance**

$$Var(x) = \int_{-\infty}^{+\infty} (x-\mu)^2 f(x) dx = \int_{1}^{3} (x-2)^2 \frac{1}{2} dx = \frac{(x-2)^3}{6} = \frac{1}{6} + \frac{1}{6} = \frac{1}{3}$$

- **Standard deviation**

$$\sigma = \frac{1}{\sqrt{3}}$$

- **Coefficient of Variation**

$$C.O.V. = \frac{StandardDeviation}{Mean} = \frac{\sigma}{\mu} = \frac{1}{2\sqrt{3}}$$

Covariance and correlation

Covariance

$$Cov(x) = \sigma_{xy} = E[(x-\mu_x)(y-\mu_y)] = E(xy) - E(x)E(y)$$

- Independent variables => covariance is 0

$$E(xy) = E(x)E(y)$$

- Reverse not true

Correlation

- Normalized value of covariance
- Always between =1 and 1
- Dependence implies correlation
- Reverse NOT true

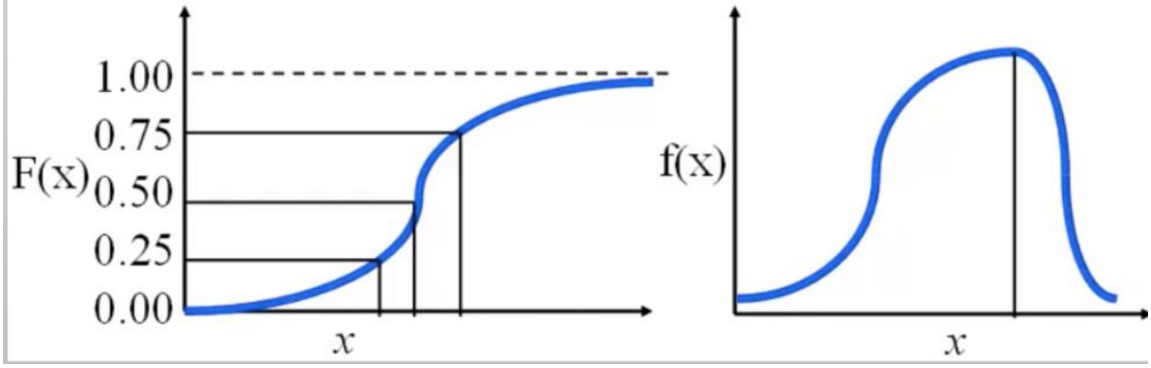
$$Correlation(x, y) = \rho_{xy} = \frac{\sigma_{xy}^2}{\sigma_x \sigma_y}$$

Quantiles, Median, Mode

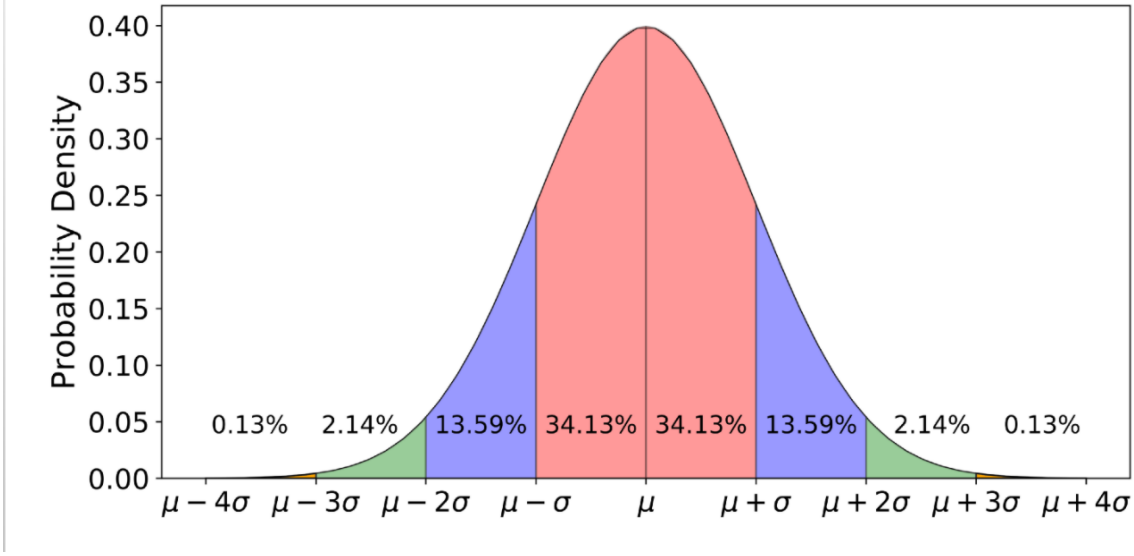
- Quantile
  - The x value at which the CDF takes a value a is called the a-quantile or 100a-percentile.

$$P(x \leq x_\alpha) = F(x_\alpha) = \alpha$$

- Median
  - 50-percentile or 0.5-quantile of a random variable
- Mode
  - The most likely value (x that has the highest probability; x at which PDF is maximum)



68-95-99.7 rule



Normal distribution

- The sum of a large number of independent observations from any distribution has a normal distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, \quad -\infty \leq x \leq +\infty$$

- Normal variate $N(\mu, \sigma)$
- Standard normal distribution $N(0, 1)$

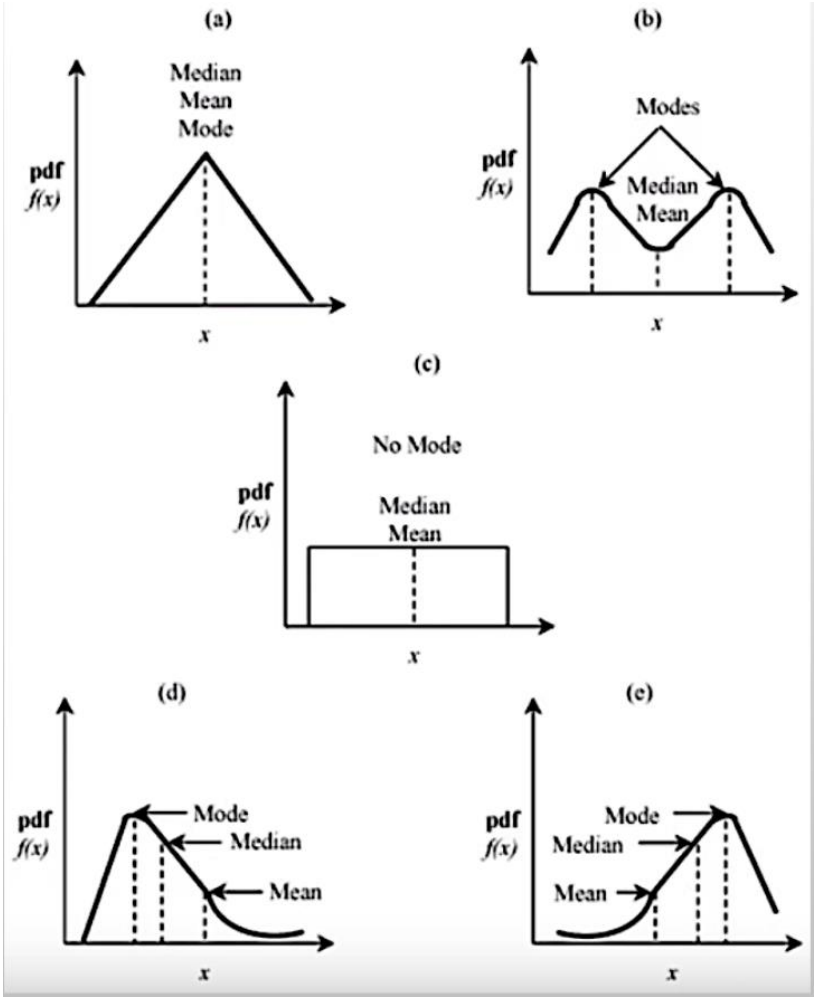
Summarising data by a single number

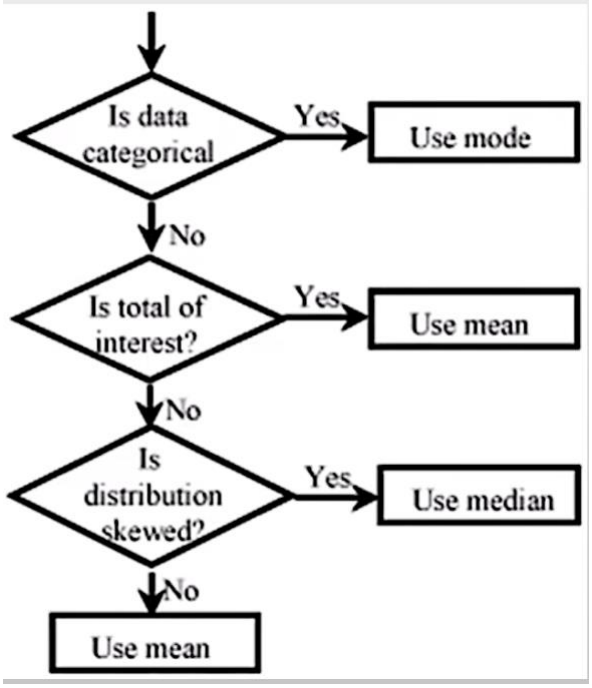
Indices of central tendencies

- Mean
- Median
- Mode

Population vs Sample

- Sample Mean
  - Sum of all observations divided by number of observations
- Sample Median
  - Sort obs in increasing order. Middle of the series. Even number mean of middle 2 values.
- Sample Mode
  - Histogram midpoint where the histogram peaks (most frequent category)


(a)
Median
Mean
Mode
pdf
f(x)
x
(b)
Modes
Median
Mean
pdf
f(x)
x
(c)
No Mode
Median
Mean
pdf
f(x)
x
(d)
Mode
Median
Mean
pdf
f(x)
x
(e)
Mode
Median
Mean
pdf
f(x)
x


Is data categorical
Yes
Use mode
No
Is total of interest?
Yes
Use mean
No
Is distribution skewed?
Yes
Use median
No
Use mean

Examples

- Most used resource in a system
- Interarrival time
- Load on a computer
- Average configuration
    - Number of devices
    - Memory size
    - Number of processors

Misuses of means

1. Using mean of significantly different values
   (10 + 1000)/2 = 505
2. Using mean without regard to the skewness of distribution

System A	System B
10	5
9	5
11	5
10	4
10	31
Sum=50	Sum=50
Mean=10	Mean=10
Typical=10	Typical=5

3. Multiplying means to get the mean of a product

   $$E(xy) \neq E(x)E(y)$$

   Ex. Time sharing system
    - Avg no of users – 23
    - Avg no of processes per user – 2
    - What is the avg no of processes?
        - is it 46?
        - No!
        - The no of processes a user spawns depends on the load on the system
4. Taking a mean of a ratio with different bases

Geometric mean

$$\dot{x} = \left(\prod_{i=1}^{n} x_i\right)^{\frac{1}{n}}$$

- Used if product of observations is of interest

Example

- Performance improvements in 7 layers

Protocol Layer	Performance Improvement
7	18%
6	13%
5	11%
4	8%
3	10%
2	28%
1	5%

- Average improvement per layer

$$\left\{(1.18)(1.13)(1.11)(1.08)(1.10)(1.28)(1.05)\right\}^{\frac{1}{7}} - 1 = 0.13$$

Examples of multiplicative metrics

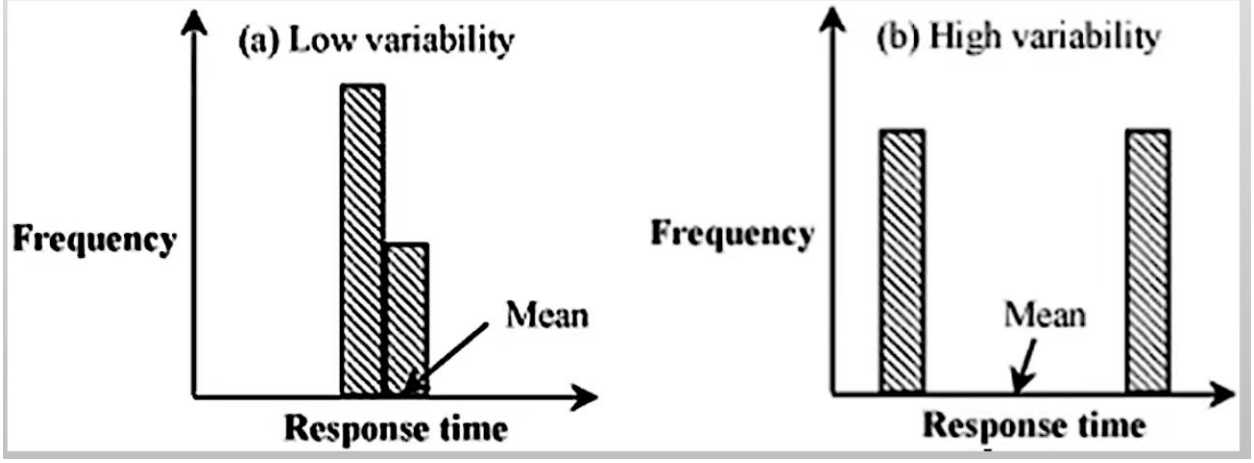
- Cache hit ratios over several levels of cache
- Cache miss ratios
- Percentage performance improvement between successive versions
- Avg error rate per hop (on a multi-hop path in a network)

Harmonic mean

$$\ddot{x} = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \ldots + \frac{1}{x_n}}$$

- Used when an arithmetic mean can be justified for $1/x_i$

Summarising variability



Indices of dispersion

- Range
  - Minimum and maximum
- Variance or standard deviation
- 10- and 90- percentiles
- Semi inter-quantile range
- Mean absolute deviation

Range

- Range = Max – Min
- Larger => higher variability

- In most cases, not very useful
    - Min often comes out to zero
    - Max often comes out to an outlier
- Unless the variable is bounded, the Max increases with the no of observations, Min decreases with the no of obs. No stable point that gives good indication of actual range.
- **Useful if there is a reason to believe that the variable is bounded**

Variance

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2, \quad where \quad \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

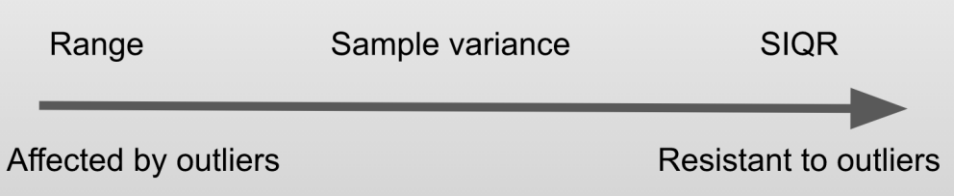
- The divisor for s2 is n-1 and not n
- The number of independent terms in a sum is also called degrees of freedom
- Expressed in units (square of units of observations)
- What is preferred to variance?
    - Variance < Std Dev < Coefficient of Variance

Percentiles

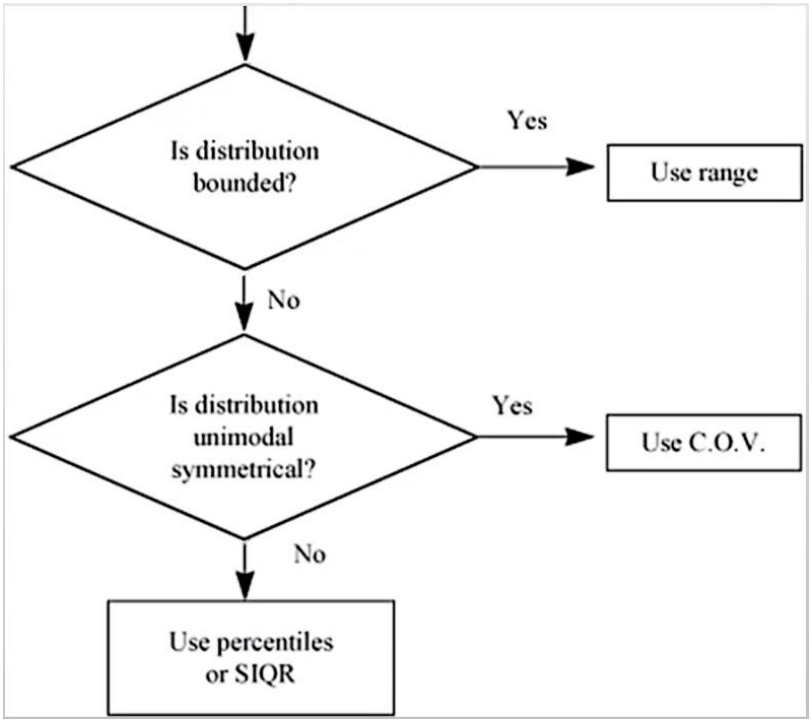
- 5-percentile and 95-percentile of a variable
    - Similar impact as Min and Max
- Can be done for any variable (even variables without bounds)
- Quantile
    - Fraction between 0 and 1
    - 0.9 quantile = 90-percentile
- Decile
    - Percentiles at multiples of 10%
- Quartile
    - Percentiles at multiples of 25%
- Semi Inter-Quartile Range

$$SIQR = \frac{Q_3 - Q_1}{2} = \frac{x_{0.75} - x_{0.25}}{2}$$

Comparation of variation measures



Selecting Index of Dispersion



- Just guidelines
- E.g.
    - Networks
        - Upper range - worse case
        - Designed to carry 95 to 99-percentile of the observed load levels
    - Computing power
    - Power supplies - sustain peak demand

Summary

- Indices of central tendencies
    - Mode
    - Median

    - Mean
        - Arithmetic, Geometric, Harmonic means
- Indices of dispersion
    - Range
    - Variance
    - Percentiles
    - Quartiles
    - SIQR